

Evaluating genomic offset predictions in a forest tree with high population genetic structure

Juliette Archambeau, Marta Benito Garzón, Marina de Miguel, Alexandre Changenet, Francesca Bagnoli, Frédéric Barraquand, Maurizio Marchi, Giovanni G. Vendramin, Stephen Cavers, Annika Perry and Santiago C. González-Martínez

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Ecology & Hydrology



Strong limitations and assumptions

- The **assumption** that populations are currently optimally adapted to their local environment is often violated in forest trees.
→ e.g., Rehfeldt *et al.* 2002, 2003, Savolainen *et al.* 2007, Pedlar & McKenney 2017, Rehfeldt *et al.* 2018, Fréjaville *et al.* 2020.
- The **assumption** that populations in similar environments share the same adaptive alleles is often violated, as phenotypes can arise from different genetic variants and evolutionary trajectories depend on the adaptive landscape.
→ Laruson *et al.* 2020, Lotterhos 2023

Genomic offset predictions have been mostly **validated** with **simulations** or **common gardens**.

Further empirical validation needed before they can **reliably** guide conservation and management!

How to account for neutral population genetic structure?

- Greater **genetic drift** (e.g., in small populations) results in greater allele turnover, and thus higher genomic offset predictions **without any selection pressure** being involved (Laruson *et al.* 2020).
- Population genetic structure can be accounted for when:
 - selecting sets of candidate SNPs.
 - estimating the genomic offset.
- **Trade-off:**
 - **over-correction** may remove adaptive signals
 - **under-correction** may confound drift with adaptation.

Study goals

- *Objective 1.* Evaluating **consistency** of genomic offset predictions.
- *Objective 2.* Evaluation **empirical validity** of genomic offset predictions in **common gardens**.
- *Objective 3.* Evaluation **empirical validity** of genomic offset predictions in **natural populations**.

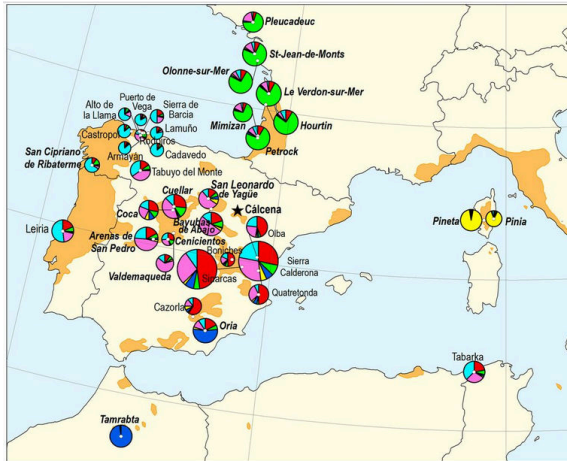
Maritime pine (*Pinus pinaster* Ait)



Caudullo *et al.* (2017)

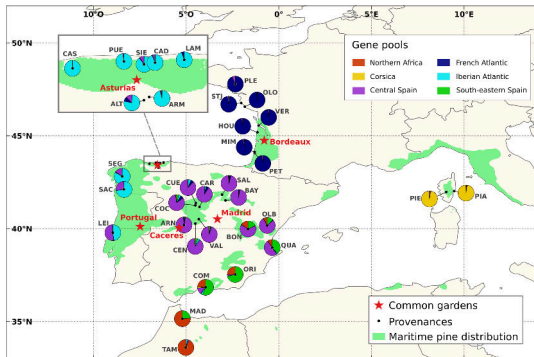
Maritime pine (*Pinus pinaster* Ait)

A strong neutral population genetic structure.



Jaramillo-Correa *et al.* (2015)

Clonal common garden network (CLONAPIN)



454 clones

34 populations



9,817 genetic markers (SNPs)
after filtering



Height and mortality data



6 climatic variables

Reference climate: 1901-1950

Future climate: 2041-2060

SSP3.7-0

5 general circulation models



Selection of the climatic variables

- Based on **biological relevance** for maritime pine
- Based on **contribution to genetic variance** (via RDA-based stepwise selection; Capblancq et al. 2021),
- Based on the **magnitude** of their **exposure to climate change**.

Label	Description	Unit
bio1	Mean annual temperature	Celsius degrees (°C)
bio3	Isothermality (bio2/bio7) ($\times 100$)	Index
bio4	Temperature seasonality (standard deviation $\times 100$)	Celsius degrees (°C)
bio12	Annual precipitation	Millimeters (mm)
bio15	Precipitation seasonality (coefficient of variation)	Index
SHM	Summer heat moisture index	°C / mm

Partitioning of the genetic variance

RDA models	Total exp. variance	Relative exp. variance	P-value
Full model: $Y \sim \text{clim} + \text{geo} + \text{pgs}$.	0.803	1.000	0.001
Pure climate model: $Y \sim \text{clim} \mid (\text{geo} + \text{pgs})$	0.079	0.098	0.036
Pure pop. gen. structure model: $Y \sim \text{pgs} \mid (\text{geo} + \text{clim})$	0.136	0.169	0.001
Pure geography model: $Y \sim \text{geo} \mid (\text{pgs} + \text{clim})$	0.083	0.103	0.002

→ **Single variables** explained **29.8%** of the variance.

→ **50.5%** of the variance **could not be uniquely attributed** to a given variable (i.e., confounded effects).

Y = genomic data (i.e. population allele frequencies filtered for MAF)

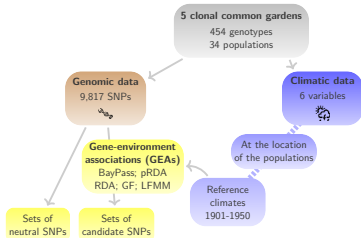
clim = climate

pgs = population neutral genetic structure (PCA main axes)

geo = geography (i.e., distance-based Moran's eigenvector maps).

Objective 1. Evaluating prediction consistency

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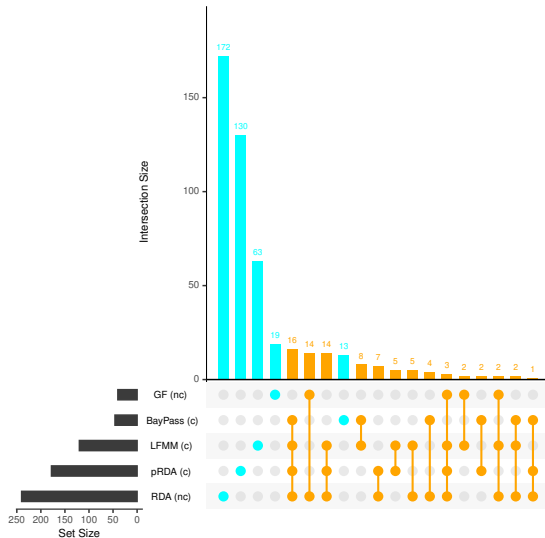


Objective 1. Evaluating prediction consistency

	RDA/pRDA	GF	BayPass	LFMM
Method name	(Partial) Redundancy Analysis	Gradient Forest	Generalized Dissimilarity Modeling	Latent Factor Mixed Model
Genomic input data	Population allele frequencies	Population allele frequencies	Population allele counts	Genotype allele counts
Multivariate X?	Yes	Yes	No	Yes
Multivariate Y?	Yes	No	No	No
Linearity	Linear	Non-linear	Linear	Linear
Population structure correction	No correction (RDA) Correction with PC scores (pRDA)	No correction	Correction with a population covariance matrix (Ω)	Correction with latent factors
Metric to evaluate the SNPs	Mahalanobis distances between each locus and the center of the RDA space along the significant axes	Empirical p -values based on $\mathcal{R}^2$ rank	Median Bayes Factors (BFs) calculated over 3 runs	Locus-specific effect sizes
Threshold to select the outlier SNPs	False discovery rate of 5%	0.5% of SNPs with lowest empirical p -values	Median Bayes factor across five runs > 10 deciban units	False discovery rate of 5%
R packages and references	<i>vegan</i> R package v2.6.4 Oksanen et al. (2022) Capblancq & Forester (2018) Capblancq et al. (2021)	<i>gradientForest</i> R package Ellis et al. (2012) Lotterhos & Whitlock (2014) Capblancq et al. (2023)	Gautier (2015) Jeffreys (1961)	<i>lea</i> R package v3.16.0 Caye et al. (2019) Frichot et al. (2013) Gain & François (2021)

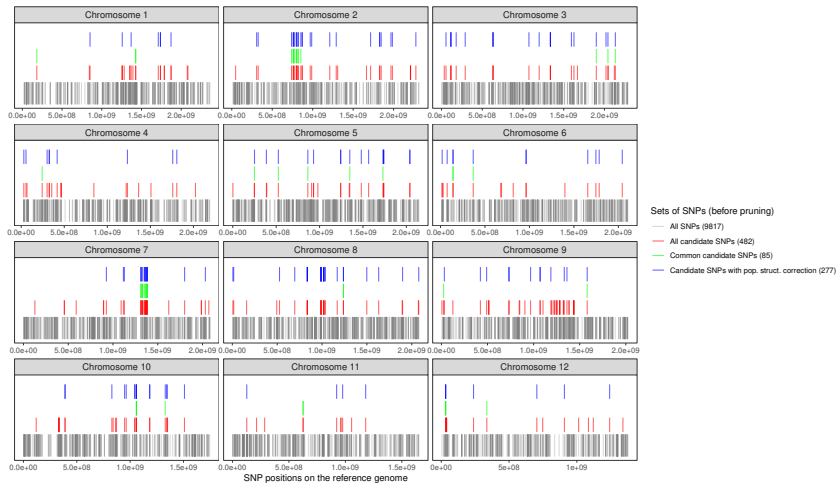
Table: Gene–environment association (GEA) methods used for identifying the SNPs potentially involved in climate adaptation.

Objective 1. Evaluating prediction consistency



Outlier SNPs identified by each of the five GEA methods.

Objective 1. Evaluating prediction consistency



SNP positions on the chromosome-scale reference genome of a related species (i.e., *Pinus tabulaeformis*; Niu et al. 2022), as such a reference is not yet available for maritime pine.

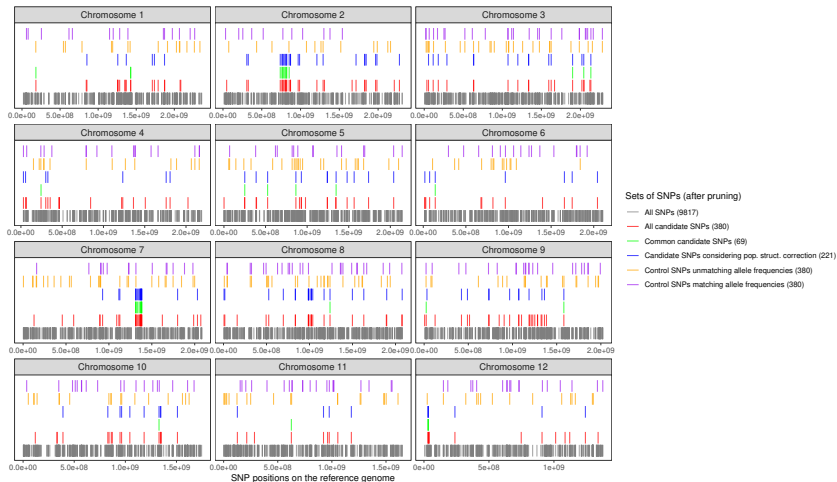
Objective 1. Evaluating prediction consistency

Why prune the candidate SNPs?

To prevent genomic regions rich in candidate SNPs from disproportionately driving the signal, assuming they are not more important than other regions.

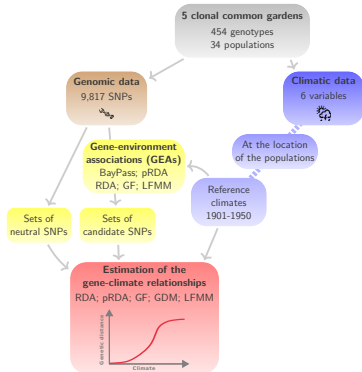
We retained one SNP per scaffold, keeping those with the lowest p -value in the RDA.

Objective 1. Evaluating prediction consistency



SNP positions on the chromosome-scale reference genome of a related species (i.e., *Pinus tabulaeformis*; Niu et al. 2022), as such a reference is not yet available for maritime pine.

Objective 1. Evaluating prediction consistency

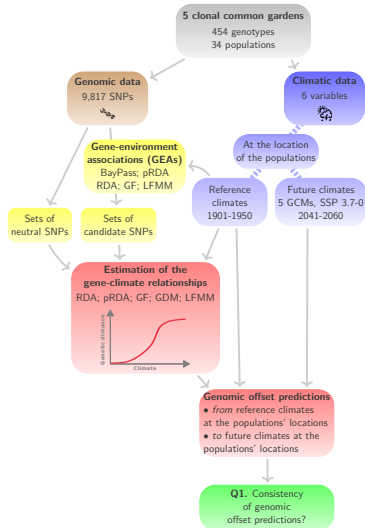


Objective 1. Evaluating prediction consistency

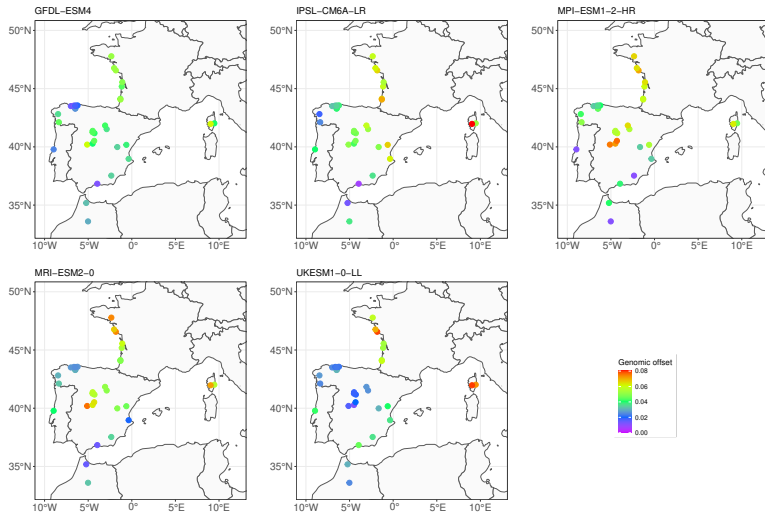
	RDA/pRDA	GF	GDM	LFMM
Method name	(Partial) Redundancy Analysis	Gradient Forest	Generalized Dissimilarity Modeling	Latent Factor Mixed Model
Genomic input data	Population allele frequencies	Population allele frequencies	F_{ST} matrices	Genotype allele counts
Multivariate X?	Yes	Yes	Yes	Yes
Multivariate Y?	Yes	No	No	Yes
Gene-climate relationships	Linear	Non-linear turnover functions	Non-linear l-splines	Linear
Population structure correction	No correction (RDA) Correction with PC scores (pRDA)	No correction	Correction with geographical distances	Correction with latent factors
R packages and references	<i>vegan</i> R package v2.6.4 Oksanen et al. (2022) Capblancq & Forester (2018) Capblancq et al. (2021)	<i>gradientForest</i> R package Ellis et al. (2012) Lotterhos & Whitlock (2014) Capblancq et al. (2023) Fitzpatrick et al. (2021)	<i>gdm</i> R package v1.5.0.9.1 Fitzpatrick (2025) Ferrier et al. (2007) Fitzpatrick et al. (2015)	<i>lea</i> R package v3.16.0 Caye et al. (2019) Frichot et al. (2013) Gain & François (2021)

Table: Genomic offset methods used in the present study.

Objective 1. Evaluating prediction consistency

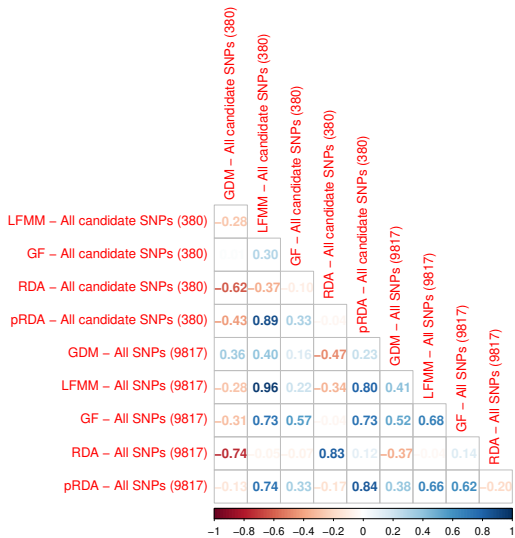


Objective 1. Evaluating prediction consistency

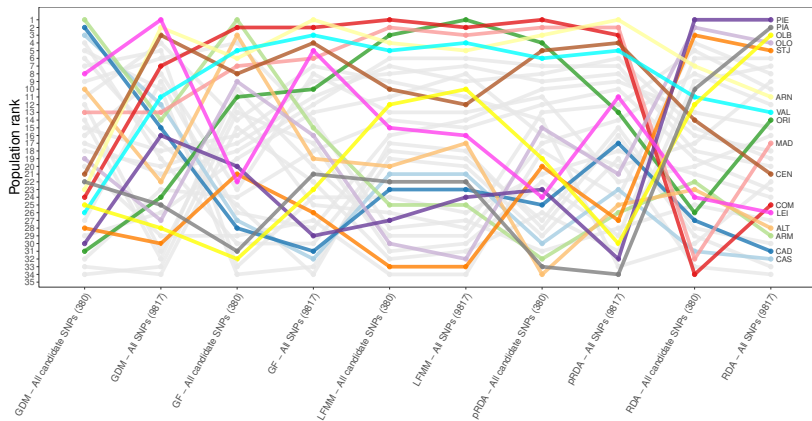


Genomic offset predictions based on RDA and all candidate SNPs.

Objective 1. Evaluating prediction consistency

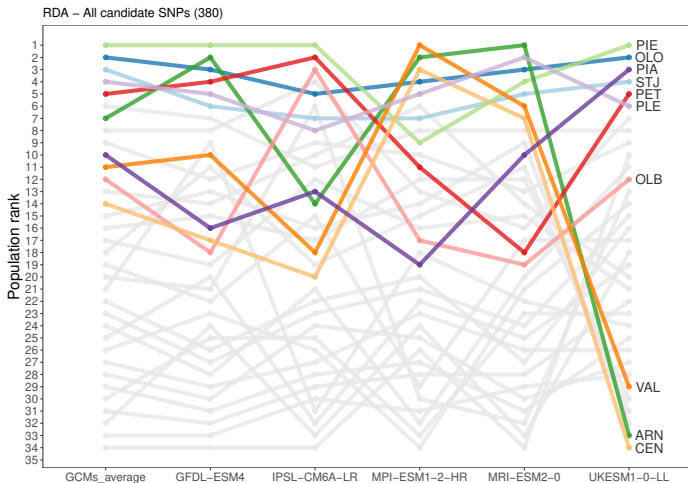


Objective 1. Evaluating prediction consistency



→ Huge variability among genomic offset predictions from different methods and SNP sets.

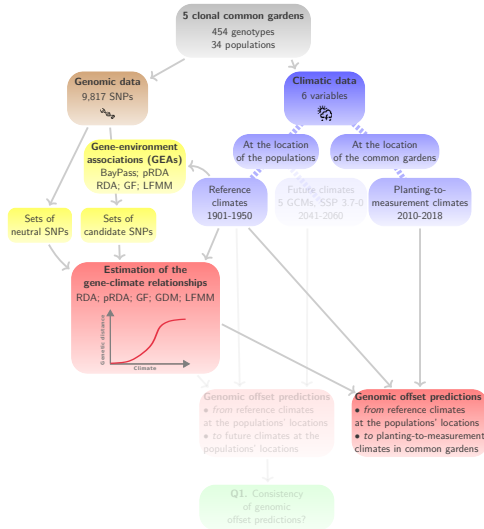
Objective 1. Evaluating prediction consistency



→ Variability among genomic offset predictions from different GCMs.

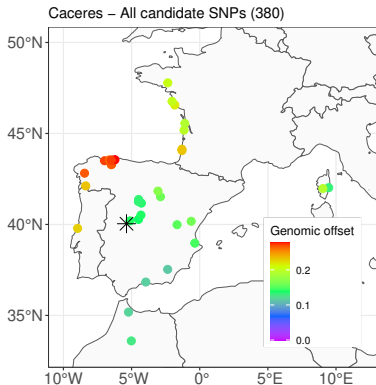
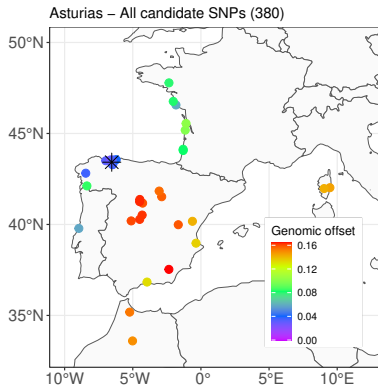
Objective 2. Evaluating empirical validity in common gardens

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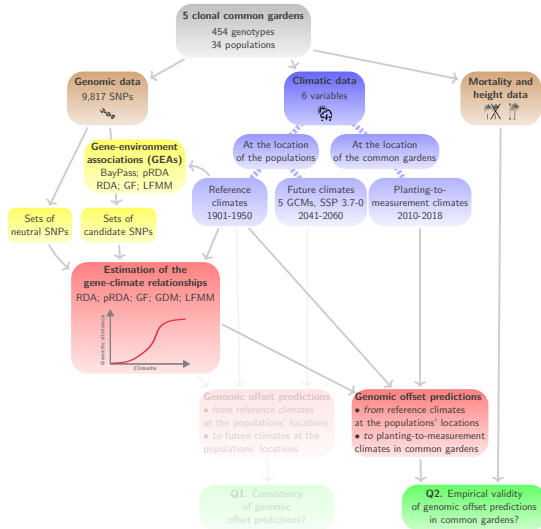


Objective 2. Evaluating empirical validity in common gardens

Genomic offset predictions based on Gradient Forest and all candidate SNPs.



Objective 2. Evaluating empirical validity in common gardens



Objective 2. Evaluating empirical validity in common gardens

We modeled the **probability of mortality** p_p in the population p independently in the five common gardens.

$$a_p \sim \text{Binomial}(N_p, p_p)$$
$$\text{logit}(p_p) = \beta_0 + \beta_H H_p + \beta_X X_p$$

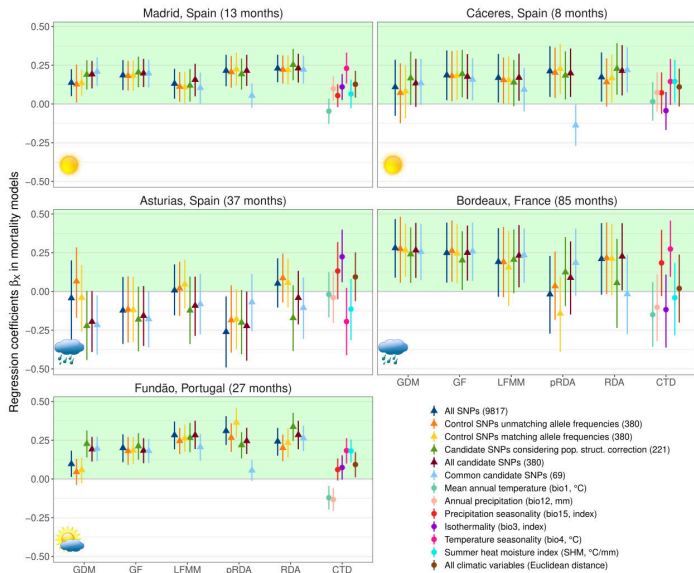
- a_p , count of individuals that died in the population p ,
- N_p , total number of individuals initially planted in the population p ,
- X_p , genomic offset predicted in the population p ,
- H_p , proxy of the mean initial tree height of the population at the planting date,
- $\mathcal{N}(0, 5)$ priors for β_0 , β_H and β_X .

Objective 2. Evaluating empirical validity in common gardens

Evaluation metrics.

- Predictive ability of genomic offset predictions evaluated based on the **strength** and **sign** of their regression coefficients β_X .
- We expected these coefficients to be **positive**, indicating that populations with **higher genomic offset** are more likely to be **climatically maladapted** and, therefore, should experience **higher mortality rates**.
- These coefficients have to be **statistically unlikely** to occur by chance.

Objective 2. Evaluating empirical validity in common gardens



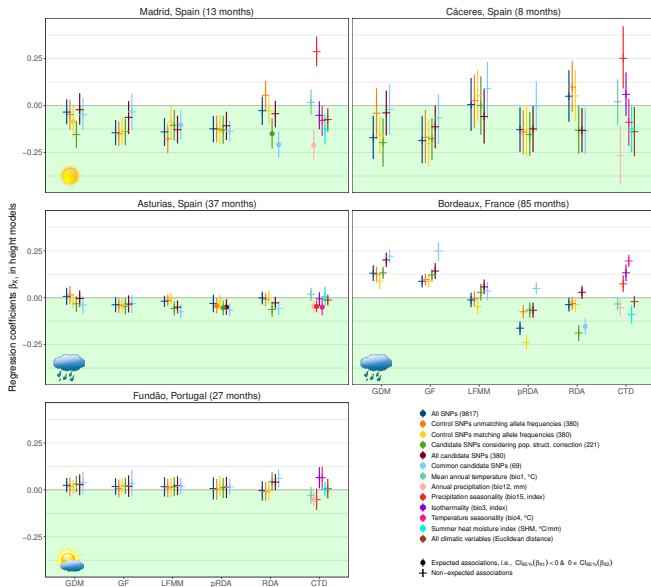
Objective 2. Evaluating empirical validity in common gardens

We modeled **tree height** independently in the five common gardens

$$Y_{ipb} \sim \mathcal{N}(\mu_p, \sigma^2)$$
$$\mu_p = B_b + \beta_{X1}X_p + \beta_{X2}X_p^2 + \beta_H H_p$$

- Y_{ipb} , height of the individual i in the population p and the block b ,
- B_b , block intercepts,
- X_p , genomic offset predicted in the population p ,
- H_p , proxy of the mean initial tree height of the population at the planting date,
- σ^2 , residual variance,
- $\mathcal{N}(0, 1)$ prior for B_b , β_{X1} , β_{X2} and β_H ,
- $\text{Exp}(1)$ prior for σ .

Objective 2. Evaluating empirical validity in common gardens



Objective 2. Evaluating empirical validity in common gardens

Why not use correlation coefficients to compare the models?

This would require:

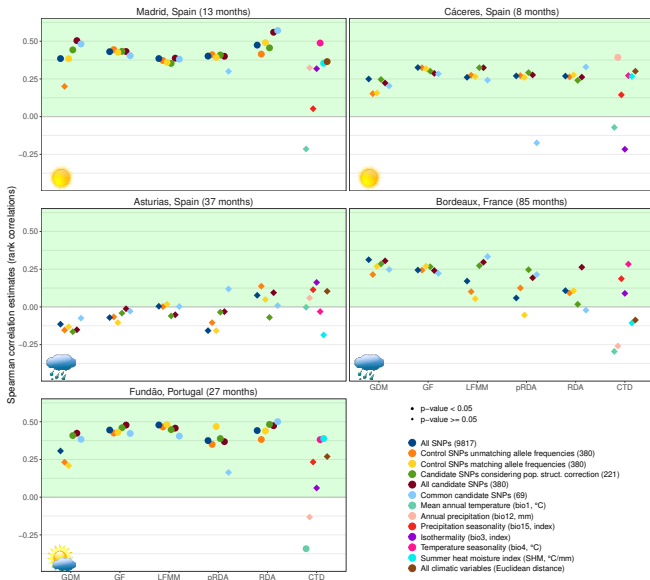
1. **Extracting residuals** of a model predicting height and mortality using only the confounding factors.
2. **Estimating the correlations** between these residuals and the genomic offset predictions.

This two-step approach has two main drawbacks:

- Does not allow for modeling higher-order relationships (e.g., quadratic associations).
- **Does not propagate the uncertainty in parameter estimates.**

See [Freckleton 2002](#) and [García-Berthou 2001](#) for critiques of two-step processes using residuals from previous models; and see [Lotterhos 2024](#) for a discussion on evaluation metrics specific to the genomic offset framework.

Objective 2. Evaluating empirical validity in common gardens

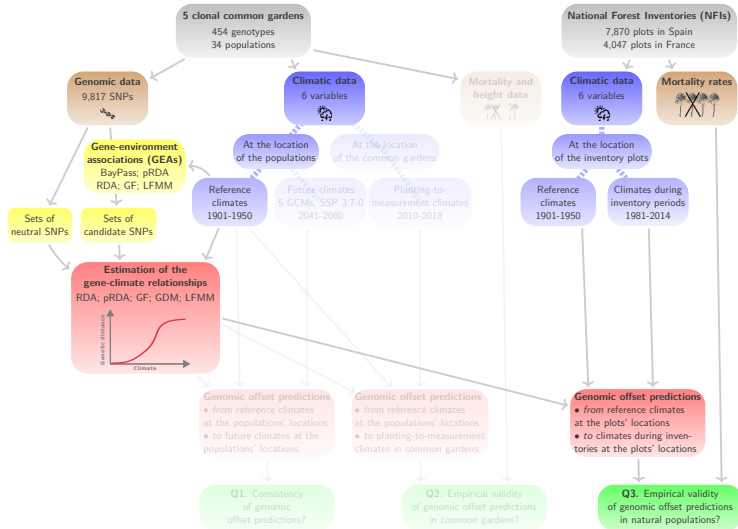


Objective 2. Evaluating empirical validity in common gardens

- **Validation** of genomic offset predictions with **mortality** data, especially in common gardens experiencing **intense summer droughts**.
- No single method or SNP set consistently outperformed the others.
- **Lack of validation** with **height** data, potentially due to **strong population genetic structure** in maritime pine, likely confounded with gradients of climatic adaptation.
- Raises concerns about the **reliability** of **height** as a **fitness proxy** for maritime pine.

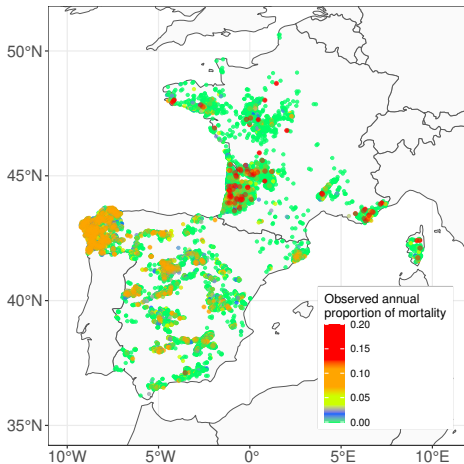
Objective 3. Evaluating empirical validity in natural populations

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Objective 3. Evaluating empirical validity in natural populations

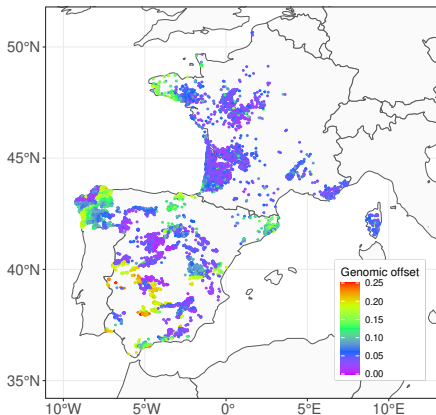
National Forest inventory (NFI) data: 11,917 plots in Spain and France.



Objective 3. Evaluating empirical validity in natural populations

Genomic offset predictions at the location of the inventory plots:

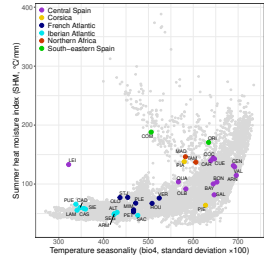
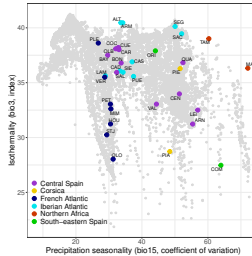
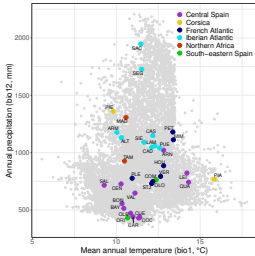
- Reference climate: 1901-1950.
- Climate of the inventory period for the predictions.



E.g., Predictions with
GF and all candidate
SNPs (380 SNPs)

Objective 3. Evaluating empirical validity in natural populations

Climatic coverage of the NFI plots.



→ Climatic space mostly covered, except for locations with high SHM or low *bio15* (precipitation seasonality), *bio3* (isothermality) and *bio1* (mean annual temperature).

Objective 3. Evaluating empirical validity in natural populations

We modeled the **probability of mortality** p_i of maritime pine in the plot i during the census interval Δ_i with a complementary log-log (cloglog) link:

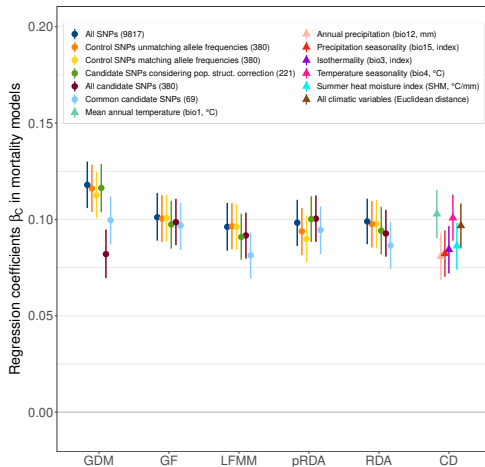
$$m_i \sim \text{Binomial}(N_i, p_i)$$

$$\log(-\log(1 - p_i)) = \beta_{0,c} + \beta_C C_i + \beta_{DBH} DBH_i + \beta_{C*DBH} C_i DBH_i + \beta_X X_i + \log(\Delta_i)$$

- m_i , number of maritime pines that died during the census interval Δ_i in the plot i ,
- N_i , total number of maritime pines in the plot i ,
- C_i , basal area of all tree species found in the plot i (to account for the competition-related mortality),
- DBH_i is the mean diameter at breast height of all maritime pines in the plot i (to account for age-related mortality),
- X_i , genomic offset predicted in the plot i ,
- $\beta_{0,c}$ are country-specific intercepts that account for the methodological differences between the French and Spanish inventories,
- $\mathcal{N}(0, 1)$ prior for $\beta_{0,c}$, β_C , β_{DBH} , β_{C*DBH} and β_X .

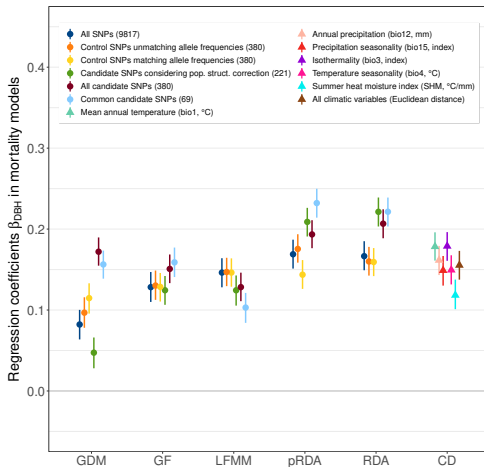
Objective 3. Evaluating empirical validity in natural populations

As expected, mortality rates positively associated with **basal area** of all tree species in each plot.



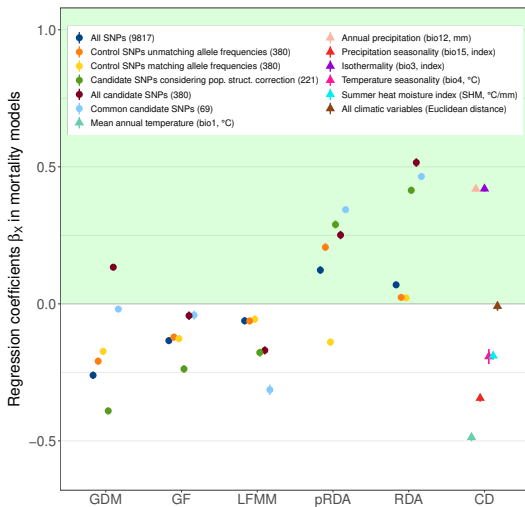
Objective 3. Evaluating empirical validity in natural populations

As expected, mortality rates positively associated with **mean DBH** of maritime pines in each plot.



Objective 3. Evaluating empirical validity in natural populations

Associations between **genomic offset predictions** and mortality rates.

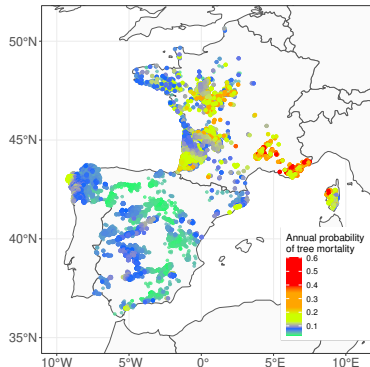
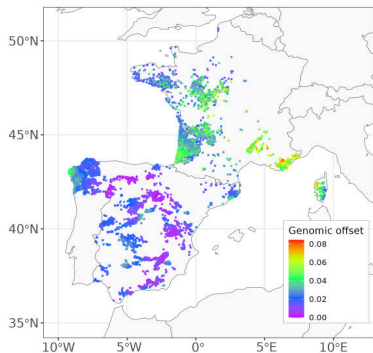


Objective 3. Evaluating empirical validity in natural populations

An example of how genomic offset predictions translate to mortality predictions.

Objective 3. Evaluating empirical validity in natural populations

An example of how genomic offset predictions translate to mortality predictions.



Genomic offset predictions based on RDA and all candidate SNPs.

Objective 3. Evaluating empirical validity in natural populations

- Associations between genomic offset predictions and mortality rates **vary** across **methods** and **SNP sets**.
- Mortality rates were **positively** associated (i.e., **expected direction**) with genomic offset predictions from **RDA** and **pRDA**, but were negatively associated with predictions from GF, LFMM and GDM.
- Regression coefficients from RDA and pRDA were **highly unlikely to occur by chance**.
- Inventory data is **inherently noisy**, making it challenging to **isolate** climate-related mortality (importance to account for confounding effects!), but may provide a **more realistic** representation of forest ecosystem **complexity**.

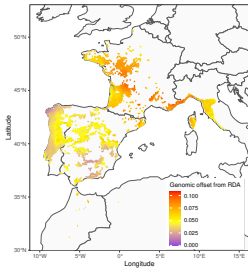
Main conclusion

Current genomic offset approaches produce **highly variable outcomes** depending on **input data** and **methodological choices**!

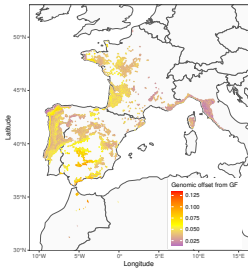
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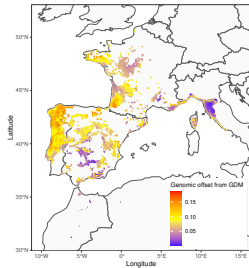
RDA



GF



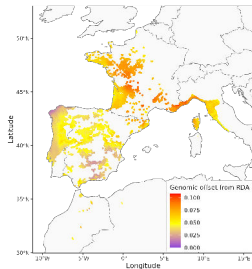
GDM



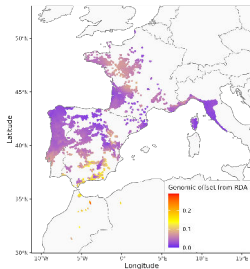
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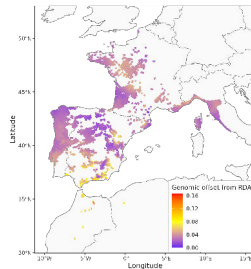
All candidates



Common candidates

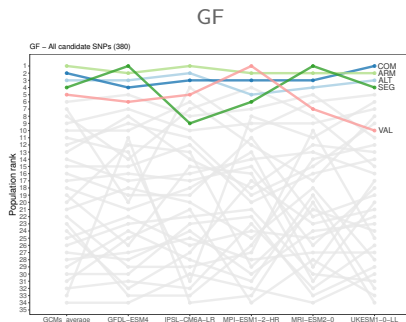
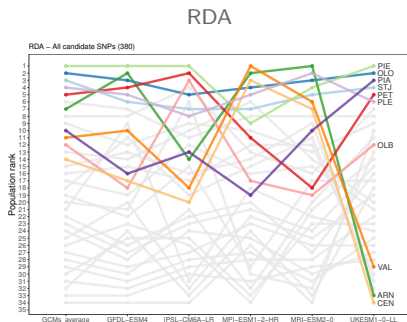


Candidate SNPs corrected for pop. structure



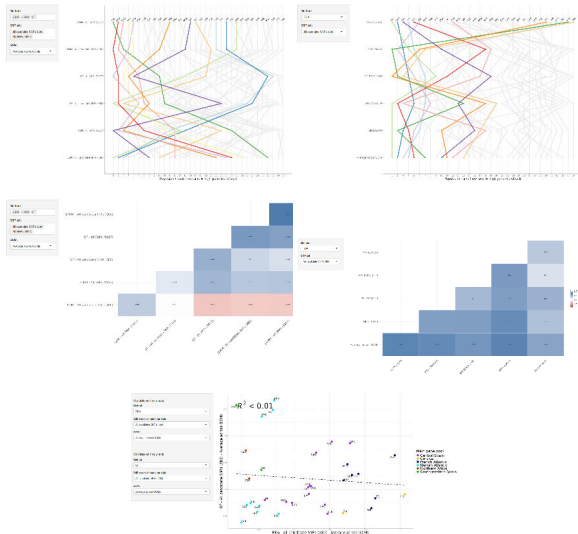
Main conclusion

Current genomic offset approaches produce **highly variable outcomes** depending on **input data** and **methodological choices**!



Visualizing uncertainty with Shiny

<https://juliettearchambeau.shinyapps.io/GenomicOffsetPredictionVariabilityInMaritimePine/>



Paper soon available in *The American Naturalist*

Github repository:

<https://github.com/JulietteArchambeau/GOPredEvalPinpin>

Dryad repository:

<https://datadryad.org/dataset/doi:10.5061/dryad.bnzs7h4jt>